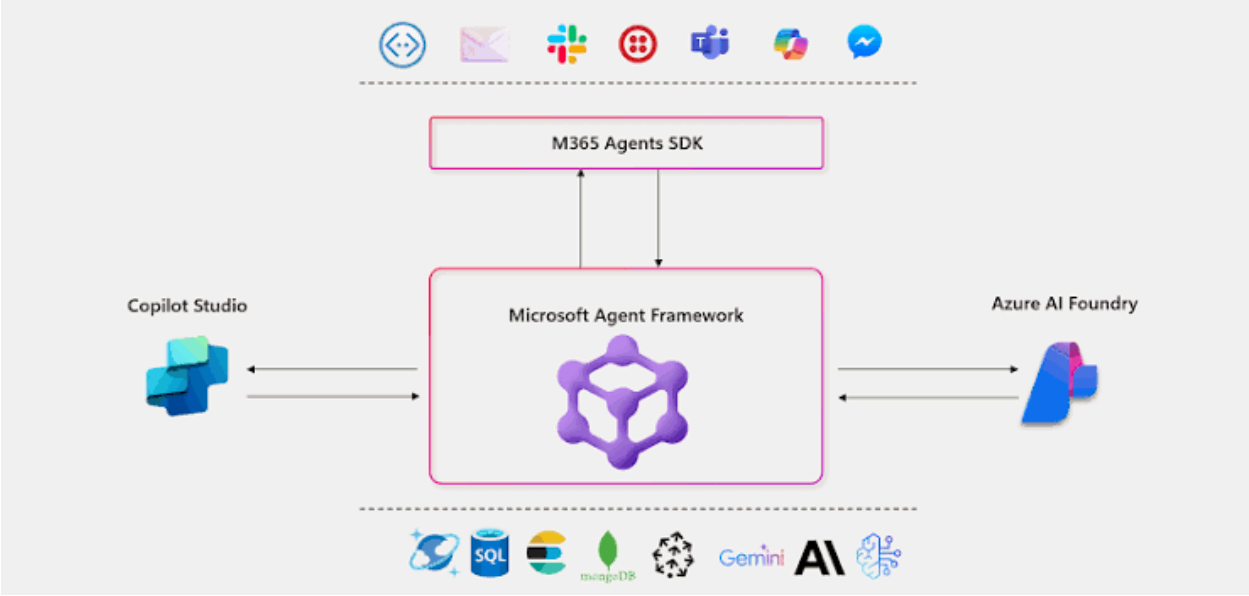
1. What is Microsoft Agent Framework(MAF)?

Microsoft Agent Framework **(MAF)** is an open-source SDK and runtime developed by Microsoft to build, orchestrate, and deploy AI agents and multi-agent workflows.



What It Does

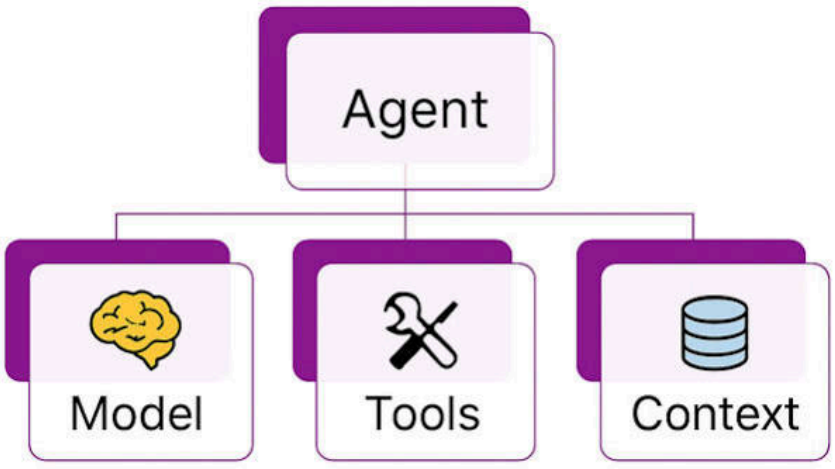
- **Unifies Past Frameworks:** It merges the enterprise-ready foundations of **Semantic Kernel** (known for production control and reliability) with the multi-agent orchestration and research patterns of **AutoGen** (known for rapid prototyping and agent collaboration).
- **Multi-Language Support:** Provides official SDKs and packages for **Python**, **.NET**, and **Go**.
- **Workflows & Deterministic Control:** Beyond free-form prompting, it features a Workflow API that lets developers design deterministic execution paths (like

Directed Acyclic Graphs or DAGs), human-in-the-loop approvals, and state checkpointing.

- **Open Standards & Interoperability:** Supports standards like the Model Context Protocol (MCP) and Agent-to-Agent (A2A) communication, and integrates smoothly with platforms like **Microsoft Foundry**, Azure OpenAI, OpenAI, and GitHub Copilot.
- **Production Observability:** Built with **OpenTelemetry** support for tracing agent reasoning, data flow, and performance monitoring.

Use Case: AI-Powered E-Commerce Multi-Agent Assistant using Microsoft Agent Framework (MAF)

Imagine building an online shopping assistant like Amazon that uses multiple AI agents to help customers purchase products.



I

Real-world scenario

A customer asks:

"I need a laptop under ₹60,000 with 16GB RAM. Find the best option, check its availability, and help me place an order."

Instead of one AI agent handling everything, MAF allows you to build multiple specialized agents that collaborate.

Why use MAF here?

Without a multi-agent framework, you might build one large agent responsible for searching products, checking stock, processing payments, and managing orders.

With MAF, you can divide responsibilities among specialized agents and coordinate their execution.

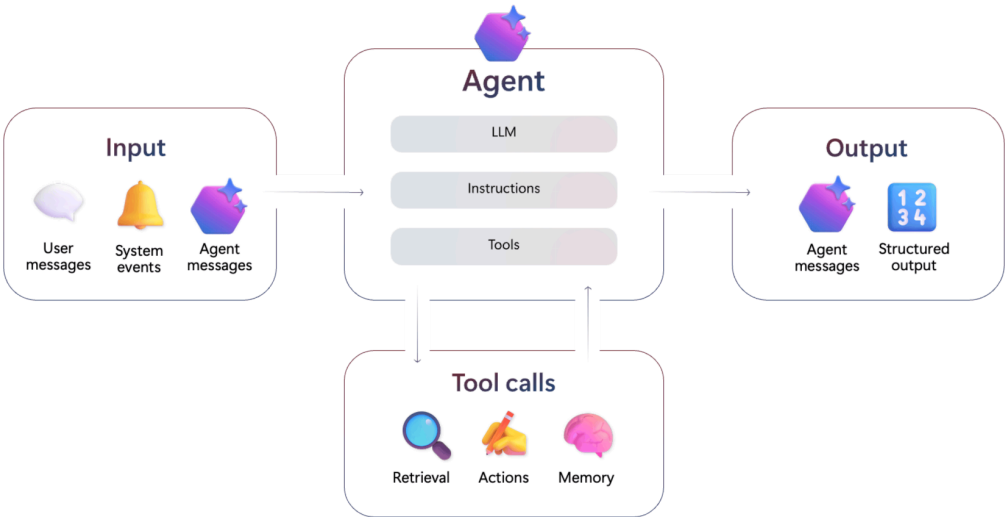
The main advantage: Each agent has a specific responsibility, making the system easier to organize, maintain, and extend.

2. What is Microsoft Foundry Agent Service?

Microsoft Foundry Agent Service is a managed cloud service in Microsoft Azure that allows developers to build, deploy, host, and manage AI agents without having to manage all the underlying infrastructure themselves.

- Microsoft Agent Framework (MAF) helps you build AI agents and define how they work together.

- Microsoft Foundry Agent Service provides the cloud environment to deploy, run, and manage those agents.



Think of MAF as the development framework and Foundry Agent Service as the hosting and management platform.

1. What does it provide?

Feature	Purpose
Agent hosting	Runs your AI agents in Azure
Agent orchestration	Supports multi-agent workflows
Model integration	Connects agents to supported AI models
Tool integration	Provides web search, file search, code interpreter, MCP and custom tools
Memory	Supports persistent conversation context
Security	Provides authentication, permissions and access control
Monitoring	Tracks agent execution, performance and errors

Feature	Purpose
Scaling	Handles changing workloads

These capabilities are documented by Microsoft.



1. Write your Python agent code using MAF.
2. Define specialized agents and their tools.
3. Test the agents locally.
4. Deploy the code to Foundry Agent Service.
5. Foundry hosts and runs the agents.
6. Connect your React or other application to the deployed agent endpoint.